Testing Representation Geometry as an Account of Distance-Based Reliability: A Preregistered Causal Evaluation in a Frozen Dermatology Foundation Model

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***Abstract*—**
Distance-based estimators are widely used to flag domain-shifted inputs before a clinical model acts on them. Two observations motivate this study. On convolutional representations, a linear probe recovered acquisition-domain identity while distance-based estimators on the same representations performed near chance—a dissociation between decodable and accessible information. Separately, across frozen vision foundation-model backbones, local intrinsic dimensionality (LID) and within-class spectral decay were found to track how well such estimators perform. Whether the second explains the first has not been tested: the reported association is correlational across backbones, and does not establish that intervening on geometry within a backbone changes what these estimators recover. We evaluated that hypothesis on a frozen dermatology foundation model (PanDerm ViT-L/16), with ISIC 2019 as the in-distribution population and PAD-UFES-20 as the domain-shifted population; neither dataset formed part of the backbone's pretraining corpus. Rather than proposing a new detector, we assessed an explanatory hypothesis under a preregistered protocol in which claims, statistical tests, and multiplicity correction were fixed before execution. Representations were reshaped post hoc along an intervention-strength ladder against a matched-budget fine-tuning control. An implementation-integrity gate confirmed that domain information and in-distribution task performance persisted after intervention, and a manipulation check confirmed that the targeted geometry was measurably altered before any causal comparison was interpreted. Four estimators—Mahalanobis, cosine-to-centroid, k-nearest-neighbour, and kernel density—were evaluated across five seeds with Holm–Bonferroni correction. The probe–reliability dissociation remained statistically detectable but was approximately two orders of magnitude smaller than originally reported (mean AUROC gap $\approx$0.002 versus $\approx$0.3), with estimators already near ceiling on the unmodified representation. LID and spectral-decay slope showed no Holm-significant association with estimator performance; condition number was consistently associated with all four. A comparison matching geometry across differing nominal intervention strengths yielded no Holm-significant difference in degradation. These results provide limited empirical support for a causal explanatory reading of the geometry–performance association under the evaluated backbone, layer, and task, and indicate that associations observed across representations warrant direct intervention-based evaluation before being interpreted as explanatory.

***Index Terms*—Out-of-distribution detection, reliability estimation, representation geometry, foundation models, causal evaluation, preregistration, dermatology.**

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I. Introduction

Automated diagnostic systems deployed in dermatology routinely encounter images acquired under conditions their training data does not represent: different capture devices, illumination, populations, and clinical settings. Because a classifier's confidence is unreliable on such inputs, deployed systems commonly pair the classifier with a separate safeguard that estimates whether an input falls outside the training distribution. Distance-based estimators—Mahalanobis distance to class centroids and its relatives—remain the dominant choice for this role, and they rest on a geometric premise: that inputs from an unseen acquisition domain occupy a measurably different region of the representation space than the training population does.

Two lines of evidence complicate that premise, and they have so far remained separate.

The first concerns whether the premise holds at all. On convolutional representations of skin lesions, a linear probe recovered acquisition-domain identity at high accuracy—establishing that the domain signal was present and linearly decodable—while distance-based estimators operating on the same representations performed near chance [1]. The information distinguishing the domains was decodable, but not accessible to the estimator intended to exploit it. This dissociation is clinically relevant because it describes a failure mode in which the safeguard reports normality precisely where a probe could establish that the input is anomalous.

The second concerns what governs estimator performance. Across a range of frozen vision foundation-model backbones, two properties of the in-distribution representation—its local intrinsic dimensionality (LID) and the decay profile of its within-class covariance spectrum—were found to track how well Mahalanobis-style estimators perform [2]. That work reports an association measured across representations produced by different pretraining regimes, and characterizes it as tracking rather than causation.

Whether the second line of evidence explains the first is an open question, and it is the question we take up. An association observed across representations is consistent with several readings: geometry may govern what distance-based estimators can recover; or both geometry and estimator performance may follow jointly from properties of the pretraining regime that distinguish one backbone from another. These readings are

not separable by comparing backbones, because backbones differ in many respects at once. Distinguishing them requires holding the representation's provenance fixed and intervening on geometry directly—and if geometry governs accessibility in the sense the dissociation would require, such an intervention should change what distance-based estimators recover from a representation whose domain content is otherwise preserved.

We evaluated that hypothesis on a frozen dermatology foundation model, within a single fixed backbone rather than across backbones. Rather than proposing a new out-of-distribution detector, we assessed an explanatory hypothesis, under a preregistered protocol incorporating implementation-integrity verification, manipulation checking, and prespecified statistical analysis with multiplicity correction fixed in advance of execution. Only checkpoints clearing both verification stages enter the causal comparison.

Our findings provide limited empirical support for a causal explanatory reading of the association, under the evaluated backbone, layer, and task. The probe–reliability dissociation remained statistically detectable, but at approximately two orders of magnitude smaller a margin than originally reported, with distance-based estimators already operating near ceiling on the unmodified representation. The two preregistered geometry metrics showed no Holm-significant association with estimator performance. These results indicate that associations observed across representations warrant direct intervention-based evaluation before being interpreted as explanatory.

The contributions of this work are as follows.

- We present a preregistered evaluation protocol for testing whether an observed representation-level association supports a causal explanatory reading, incorporating implementation-integrity gating, explicit manipulation checking, and multiplicity correction fixed before execution.
- We formulate and test the hypothesis that links two previously separate observations—the probe–reliability dissociation and the geometry–performance association—by intervening on representation geometry within a single fixed frozen backbone, rather than comparing geometry across backbones that differ in many respects at once.
- We characterize the extent to which the previously reported probe–reliability dissociation is observable in this regime, and the degree of empirical support the explanatory hypothesis receives under the evaluated backbone, layer, and task.

II. Related Work

A. Distance-Based Reliability Estimation

Distance-based out-of-distribution detection scores an input by its position relative to the training distribution in representation space. The Mahalanobis detector fits class-conditional Gaussians to training features and scores an input by its minimum Mahalanobis distance to any class centroid [3]; nearest-neighbour, kernel-density, and cosine-to-centroid variants replace the parametric fit with local or directional statistics but retain the same premise, that distributional novelty manifests as geometric displacement. The family is well suited to clinical deployment because it requires no outlier data at training time and attaches to an existing classifier without retraining [3]. The premise is nonetheless an assumption about representation geometry, and this paper concerns the conditions under which it holds.

B. Decodability Versus Accessibility

Linear probing trains a classifier independently of the model in order to measure what information a representation linearly encodes [4]. The distinction matters here because a probe and a distance-based estimator can disagree: a probe may recover acquisition-domain identity from a representation on which distance-based scores are uninformative. Prior work on convolutional networks reported exactly this dissociation for skin-lesion representations: probes recovered acquisition-domain identity at 0.72–0.81 AUROC while Mahalanobis-family scores sat near 0.40 under the standard higher-distance-is-more-OOD convention [1]. That report is explicit that the scores did not approach chance but were directionally misaligned: their orientation-free separability was 0.58–0.60, so the ordering retained signal while pointing the wrong way. That report motivates the present evaluation; whether it describes the frozen foundation-model regime is the question we examine.

C. Representation Geometry and Estimator Performance

Estimator performance has been linked to two properties of the in-distribution representation: its local intrinsic dimensionality, estimated by the Levina–Bickel maximum-likelihood estimator [5], and the decay profile of its within-class covariance spectrum [2]. That work evaluates a range of frozen vision foundation-model backbones and reports that these two quantities track how Mahalanobis-style detector performance varies across them. The relationship is characterized as tracking rather than causation, and the study does not employ probing classifiers or examine the dissociation described above. Its interventions target an internal term of its own derivation rather than detector performance itself.

A substantial body of work manipulates related geometric quantities directly, mostly outside out-of-distribution detection. Post-hoc whitening and isotropy-oriented transforms reshape the covariance spectrum of frozen embeddings, either by removing dominant principal directions [6], by full or partial decorrelation [7], [8], or by graded per-dimension spectral reweighting [9]. Evidence on semantic cost is mixed: partial-strength variants report downstream gains, while a systematic evaluation of five full-strength whitening variants reports degradation on classification tasks [10]. A separate line targets geometry during training rather than post hoc, through spectral or intrinsic-dimensionality regularizers [11], [12]. We draw the geometry-manipulation operator used in our causal comparison from the post-hoc, dose-controllable end of this literature, for reasons detailed in Section III.

D. Frozen Foundation Models and Cross-Backbone Comparison

Dermatological image analysis has moved toward large pretrained encoders used as frozen feature extractors, with

lightweight task-specific components trained on top [13]. In frozen foundation models, the representation geometry is primarily inherited from large-scale pretraining rather than optimized end-to-end for the downstream clinical task. Comparing geometry and detector performance across such backbones establishes that the two covary, but backbones differ simultaneously in pretraining corpus, objective, architecture, and scale, so an association measured across them cannot separate a geometric explanation from a shared dependence on pretraining. Distinguishing these readings requires holding provenance fixed and intervening on geometry directly. Our contribution is an evaluation of that kind, in the same spirit as validation studies that test whether a reported effect survives a change of setting, applied here to the explanatory status of a reported association rather than to its reproducibility.

III. Methods

A. Backbone and Populations

All experiments use a single frozen pretrained dermatology foundation model, PanDerm ViT-L/16 [13], as a feature extractor. The backbone is never updated: every intervention described below operates on its output embeddings, which are 1024-dimensional. Embeddings were extracted once under a fixed preprocessing pipeline and stored as an immutable artifact with checksum verification, so that all downstream analyses share bit-identical inputs.

The in-distribution population is ISIC 2019 [14], [15], [16], comprising 25,331 dermoscopic images across eight diagnostic classes. We partition it with a fixed seed into a held-out evaluation set (20%, $n = 5{,}067$), a validation set (20% of the remainder, $n = 4{,}053$), and a training population ($n = 16{,}211$) used to fit all distribution-dependent quantities. The validation split is not used in the analyses reported here. This partition is specific to the present study and does not correspond to the challenge's own splits.

The domain-shifted population is PAD-UFES-20 [17], contributing 2,298 close-up clinical images collected in Brazil; the modality difference from ISIC's dermoscopic images accompanies differences in device, population, and clinical setting. PAD-UFES-20 is mapped into the same eight-class label space and populates six of the eight classes. The evaluation population combines held-out ISIC and PAD-UFES images, $n = 7{,}365$ in total. Class labels are used for the in-distribution task and for class-conditional estimator fitting; the domain label (ISIC versus PAD-UFES) is the target of the domain probe.

Neither ISIC 2019 nor PAD-UFES-20 forms part of the backbone's pretraining corpus, which draws on eleven separately described sources [13].

We additionally define a fixed nuisance direction $w = \mathrm{unit}(\mu_{\mathrm{ISIC}} - \mu_{\mathrm{PAD}})$, the unit vector between population means, computed once and frozen before any intervention was trained.

B. Geometry Metrics

Four geometric quantities are measured on every evaluated representation. Two are preregistered as the metrics the account under evaluation specifies; two are recorded alongside them.

Local intrinsic dimensionality (LID) is estimated per point by the Levina–Bickel maximum-likelihood estimator [5] with $k = 50$ neighbours,

$$\mathrm{LID}_k(z) = -\left[\frac{1}{k}\sum_{j=1}^{k}\log\frac{r_j(z)}{r_k(z)}\right]^{-1}, \tag{1}$$

where $r_j(z)$ is the distance from z to its j-th nearest neighbour in the same feature set, and reported as the dataset mean.

Within-class spectral-decay slope is obtained from the within-class scatter matrix $S_w = \frac{1}{N}\sum_c \sum_{i:y_i=c}(z_i-\mu_c)(z_i-\mu_c)^\top$, computed without regularization so that tail eigenvalues are not distorted. Its eigenvalues are sorted in descending order and fitted by ordinary least squares as $\log\lambda_i = a + b\,i$ over the full range $i = 1, \ldots, d$; the reported slope is $|b|$.

Condition number is computed on the fitted Mahalanobis precision matrix, and *per-direction variance* is the variance of the projection of the evaluation embeddings onto the frozen nuisance direction, $\mathrm{Var}(z^\top w)$.

C. Representation Intervention and Dose Ladder

The intervention is a bottleneck adapter applied to the frozen embedding,

$$z' = z + W_2\,\mathrm{GELU}(W_1 z), \tag{2}$$

with bottleneck width $r = 16$ and the up-projection W_2 initialized to zero, so that the adapter is exactly the identity at initialization. The Intervention arm is trained with a task loss on the eight-class diagnostic objective plus an orthogonality penalty against the frozen nuisance direction, weighted by $\lambda_{\mathrm{proj}} = 0.1$. Training uses AdamW, learning rate 10^{-3}, batch size 512, for 100 epochs, repeated across five seeds $\{42, 52, 62, 72, 82\}$. Bottleneck width and λ_{proj} were selected by a preregistered rule fixed before the canonical runs.

Intervention strength is varied post hoc rather than by retraining. For each trained adapter we compute its displacement $\Delta z = W_2\,\mathrm{GELU}(W_1 z)$ and evaluate the ladder

$$z'(\alpha) = z + \alpha\,\Delta z, \qquad \alpha \in \{0, 0.25, 0.5, 0.75, 1.0\}, \tag{3}$$

so that $\alpha = 0$ recovers the unmodified representation and $\alpha = 1$ the fully trained adapter. This isolates strength from optimization variance: every rung of a given seed's ladder derives from one optimization run.

D. Control Arms

Two controls accompany the Intervention arm. The *Conventional* arm is trained identically—same adapter, same width, same schedule, same seeds—but with the task loss alone and no orthogonality penalty, giving a matched-budget ordinary fine-tuning comparison that separates intervention *type* from intervention *budget*. The *Adaptation* arm spans three capacity rungs (linear probe with the embedding unchanged; partial fine-tuning at $r = 8$; full adapter fine-tuning at $r = 16$), all trained with the task loss alone, and supplies the additional configurations required by the correlational analysis.

E. Verification Gates

No checkpoint contributes to a reported comparison before clearing two gates specified in advance.

Gate 0 (implementation integrity) confirms that the intervention left the representation usable and the domain signal present. It requires domain probe accuracy above the majority-class baseline by a fixed margin ($> 0.688 + 0.05$) and in-distribution eight-class balanced accuracy above chance by a fixed margin ($> 0.125 + 0.10$). Both probes are ℓ_2-regularized logistic regressions on standardized features, averaged over five probe seeds. Thresholds were fixed before execution. A Gate 0 failure means the intervention destroyed either the signal being studied or the task itself, and is reported as inconclusive for that checkpoint rather than as evidence.

Gate 1 (manipulation check) confirms that the intervention actually moved the geometry the account concerns. It is satisfied when at least one preregistered geometry metric shows reproducible dose dependence across the ladder; the criterion makes no assumption of linearity, strict monotonicity, or convexity, and the remaining metrics are recorded rather than discarded. An unmanipulated checkpoint tests nothing, so Gate 1 failures are excluded from causal interpretation rather than counted as negative evidence.

F. Reliability Estimators

Four distance-based estimators are evaluated on every checkpoint. The Mahalanobis detector fits class-conditional means and a shared precision matrix on the adapted training population, with regularization $\varepsilon = 10^{-5}$, and scores an input by its minimum squared Mahalanobis distance to any class centroid [3]. The cosine estimator scores by maximum cosine similarity to a class centroid; the nearest-neighbour estimator uses $k = 10$; the kernel-density estimator scores by class-conditional density. All four are fitted on the adapted *training* population and evaluated on the held-out population. Performance is reported as AUROC for separating in-distribution from domain-shifted inputs, with FPR at 95% TPR recorded alongside.

These four estimators are the family the hypothesis under evaluation concerns, all of them scoring an input by its position relative to the training distribution in representation space. Logit- and energy-based detectors are not evaluated: they depend on classifier outputs rather than on representation geometry alone, and so fall outside the hypothesis being tested. This study does not aim to rank detectors.

G. Geometry Manipulation for the Causal Comparison

Testing whether geometry, rather than nominal intervention strength, tracks estimator degradation requires an operator that moves the preregistered geometry metrics directly. The normalization-based knob originally specified for this role was found, on measurement, to move condition number substantially while leaving LID and spectral-decay slope unchanged. This is consistent with the operator's design, which adjusts feature radii while holding directions fixed [2]; both preregistered metrics are invariant to such rescaling. It was therefore not used.

The operator adopted instead is a fixed-exponent spectral tempering [9]. Given the eigendecomposition of the in-distribution training covariance with eigenvalues λ_i and eigenvectors U, embeddings are transformed as

$$z' = \mu + U \operatorname{diag}\left(\lambda_i^{-\gamma/2}\right) U^\top (z - \mu), \tag{4}$$

where $\gamma = 0$ is the identity and $\gamma = 1$ is full whitening. We use a fixed, manually specified γ rather than the adaptive per-dimension exponent of the original formulation: adaptivity was introduced there to avoid per-task tuning where no validation signal is available, whereas the present protocol supplies an independent, preregistered criterion (Gate 1) against which a prespecified γ ladder is evaluated. This operator was subjected to the same manipulation check before use, and its effect on LID and spectral-decay slope is reported in Section IV.

H. Scorer-Relevant Geometry: What the Operator Can and Cannot Move

A geometry metric is interpretable as a predictor of a scorer's ranking only if the operator that moves the metric also moves the structure the scorer reads. The two are routinely conflated: a metric is computed on a representation, an AUROC is computed from a scorer applied to the same representation, and their covariation across conditions is read as evidence that the geometry explains the ranking. That inference requires a step that is rarely checked. This subsection supplies it for the operator of (4), and the answer separates the estimators of Section III-F into two classes.

Write $\Sigma = U \operatorname{diag}(\lambda_i) U^\top$ for the in-distribution training covariance and $A_\gamma = U \operatorname{diag}(\lambda_i^{-\gamma/2}) U^\top$ for the tempering map, so that (4) reads $z' = \mu + A_\gamma (z - \mu)$. Note that A_γ is symmetric and, for $\lambda_i > 0$, invertible.

Proposition 1 (Condition number is a designed target): Under (4) the transformed covariance is $\Sigma' = U \operatorname{diag}(\lambda_i^{1-\gamma}) U^\top$, and therefore

$$\kappa(\Sigma') = \kappa(\Sigma)^{\,1-\gamma}, \qquad \text{equivalently} \qquad \log \kappa(\Sigma') = (1-\gamma) \log \kappa(\Sigma). \tag{5}$$

Proof: $\Sigma' = A_\gamma \Sigma A_\gamma^\top = U \operatorname{diag}(\lambda_i^{-\gamma/2}) \operatorname{diag}(\lambda_i) \operatorname{diag}(\lambda_i^{-\gamma/2}) U^\top = U \operatorname{diag}(\lambda_i^{1-\gamma}) U^\top$. The eigenvalues of Σ' are $\lambda_i^{1-\gamma}$, and $x \mapsto x^{1-\gamma}$ is increasing for $\gamma < 1$, so extremal eigenvalues map to extremal eigenvalues and $\kappa(\Sigma') = (\lambda_{\max}/\lambda_{\min})^{1-\gamma}$. ∎

The dose ladder is thus not merely correlated with the condition number; it sets it, on a log scale, by construction. $\gamma = 0$ leaves κ unchanged and $\gamma = 1$ drives it to unity. This is the strongest position a manipulation check can be in, and it is what makes the next result consequential rather than a technicality.

Proposition 2 (Mahalanobis scoring is invariant to the operator): Let the Mahalanobis parameters be refit on the tempered training population, as specified in Section III-F. With ridge $\varepsilon = 0$, the squared Mahalanobis score is preserved pointwise, $d'^2(z') = d^2(z)$ for every z, and the AUROC of that score is therefore identical for every γ.

Proof: Tempering is applied to every embedding, so class means map affinely, $\mu_c' = \mu + A_\gamma(\mu_c - \mu)$, and the pooled within-class scatter maps as $S' = A_\gamma S A_\gamma^\top$. With $\varepsilon = 0$ the refit precision is $P' = A_\gamma^{-\top} P A_\gamma^{-1}$, while residuals map as $z' - \mu_c' = A_\gamma(z - \mu_c)$. Hence

$$\begin{aligned} d'^2(z') &= (z' - \mu_c')^\top P'(z' - \mu_c') \\ &= (z - \mu_c)^\top A_\gamma^\top A_\gamma^{-\top} P A_\gamma^{-1} A_\gamma (z - \mu_c) \\ &= (z - \mu_c)^\top P (z - \mu_c) = d^2(z). \end{aligned} \tag{6}$$

Minimisation over c preserves the identity, and AUROC depends on the scores only through their ordering. ∎

Nothing in the proof is specific to tempering: it holds for any invertible linear map whose parameters are re-estimated afterwards. Tempering is simply the member of that family this protocol uses.

Propositions 1 and 2 together fix the status of one preregistered metric. Within this intervention family the condition number is moved by design while the Mahalanobis ranking cannot move at all, so no association between them is available to be found. A null result for that pair is entailed by the algebra and carries no information about whether representation geometry explains distance-based ranking in general. It is not evidence of a small effect, and it is not a shortfall of power at five seeds. We report it in Section IV as a bound on what the design can speak to, not as a finding about the account under evaluation.

The remaining estimators are not affine-invariant and are not covered by Proposition 2. Cosine similarity is computed after ℓ_2 normalisation and so depends on directions that A_γ rotates relative to the sphere; nearest-neighbour and kernel-density scores depend on Euclidean neighbourhood structure, which A_γ rescales anisotropically. For these three, condition number remains admissible as a candidate predictor, and the association reported for them is interpretable in a way the Mahalanobis association is not. Section IV reports a sweep confirming this separation on the reference embeddings: across $\gamma \in [0, 1]$ the realized condition number moves eight orders of magnitude while Mahalanobis AUROC moves 1.3×10^{-5}, against 1.5×10^{-2} for cosine and 1.6×10^{-3} for k-NN.

Two limits of this analysis are worth stating plainly. First, (5) is an identity in exact arithmetic. The smallest sample eigenvalue in these embeddings is of order 10^{-17}, so $\lambda_i^{-\gamma/2}$ on the tail of the spectrum is dominated by round-off and the recomputed condition number departs from $\kappa(\Sigma)^{1-\gamma}$ while remaining monotone in γ. We report the identity as algebra and the realized values as measurement, and do not treat them as the same quantity. Second, the ridge $\varepsilon = 10^{-5}$ used in practice breaks exact invariance, since εI does not transform with A_γ; the observed 1.3×10^{-5} drift in Mahalanobis AUROC is consistent with that residual and with floating-point error, and no claim of exact numerical constancy is made.

We do not attempt a corresponding result for local intrinsic dimensionality. LID is a property of local neighbourhood structure rather than of the global covariance spectrum, and the empirical profile reported in Section IV is non-monotonic in γ and consistent across all five seeds. A closed-form statement would be weaker than that observation and would risk presenting an assumption as a derivation. It is reported as an empirical finding and left there.

I. Preregistered Claims and Statistical Analysis

Claims, tests, and multiplicity correction were fixed before execution. We say the account is supported by *mediation* if the association between intervention and estimator degradation is carried by the geometry metrics rather than by intervention type or nominal strength.

To evaluate the correlational claim, we compute Kendall's τ between each geometry metric and each estimator's AUROC, pooling Intervention-arm strengths and Adaptation-arm rungs. Because configurations from a common seed share an optimization run, these pooled points are not fully independent, and the resulting τ values are interpreted as descriptive associations across the configuration set rather than as estimates from independent draws. The control-arm analysis below models the seed dependence explicitly, and a post-hoc sensitivity analysis reported with the results quantifies what the correlational p-values become when the seed, rather than the configuration, is taken as the unit of resampling. To evaluate the causal claim, we compare estimator degradation between a geometry-matched pair and a nominal-strength-matched pair, using a paired t-test with a Wilcoxon signed-rank test reported alongside, normality being assessed at analysis time rather than assumed. To evaluate the control-arm claim, we fit a mixed-effects model with an intervention-type-by-geometry interaction and seed as grouping factor. To evaluate probe decodability, we test probe AUROC against chance and against the corresponding Mahalanobis AUROC across seeds. Holm–Bonferroni correction is applied across the estimator-by-metric comparison families, and both corrected and uncorrected p-values are reported.

IV. Results

A. Gate Outcomes

All Intervention-arm and Conventional-arm checkpoints cleared Gate 0 at every rung of the dose ladder, confirming that domain information and in-distribution task performance survived the intervention. Gate 1 was satisfied on the Intervention arm: at least one preregistered geometry metric showed reproducible dose dependence across the ladder and across all five seeds. No checkpoint was excluded from the analyses below on gate grounds.

B. Probe Decodability and the Distance-Based Estimators

Across all ten evaluated configurations, a linear probe recovered acquisition-domain identity essentially perfectly, with mean probe AUROC ranging from 0.99999 to 1.00000. Mahalanobis AUROC on the same representations ranged from 0.9983 to 0.9984. Probe AUROC exceeded chance in every configuration ($p < 10^{-19}$ throughout), and exceeded the corresponding Mahalanobis AUROC in every configuration where the paired test was defined, with p between 2.5×10^{-5} and 8.4×10^{-8} across nine configurations; the remaining configuration (Adaptation, linear-probe rung) produced zero

TABLE I
KENDALL'S τ BETWEEN GEOMETRY METRICS AND ESTIMATOR AUROC ($n = 30$). BOLD INDICATES HOLM-CORRECTED SIGNIFICANCE ACROSS THE 16-COMPARISON FAMILY.

Metric	Mahal.	Cosine	kNN	KDE
LID	-0.24	0.01	0.19	0.10
Spectral slope	0.22	0.16	0.12	0.05
Condition number	**0.56**	**0.58**	**0.42**	**0.55**
Per-dir. variance	0.32	0.34	0.33	0.15

variance across seeds and is reported as degenerate rather than assigned a t-statistic.

The direction of the previously reported dissociation was therefore reproduced, and reproduced with high statistical confidence. Its magnitude was not. The mean probe-minus-Mahalanobis AUROC gap was 1.6×10^{-3} (range 1.567×10^{-3} to 1.666×10^{-3} across configurations), against a gap of approximately 0.3 in the convolutional setting where the dissociation was originally reported [1] — a ratio of roughly 200. Distance-based estimators were already operating near ceiling on the unmodified representation, with Reference-arm AUROC of 0.9984 (Mahalanobis), 0.9979 (nearest-neighbour), 0.9805 (cosine), and 0.8942 (kernel density).

C. Geometry–Estimator Associations

Table I reports Kendall's τ between each geometry metric and each estimator's AUROC across the 30 pooled configurations. Neither preregistered geometry metric reached Holm-corrected significance against any estimator: LID's strongest association was with Mahalanobis ($\tau = -0.24$, uncorrected $p = 0.065$), and spectral-decay slope's was also with Mahalanobis ($\tau = 0.22$, uncorrected $p = 0.088$). Condition number was associated with all four estimators at Holm-corrected significance (τ between 0.42 and 0.58, uncorrected p between 1.1×10^{-5} and 1.3×10^{-3}). Per-direction variance along the nuisance direction reached uncorrected significance against three estimators (p between 0.009 and 0.014) but did not survive correction.

Sensitivity to the unit of resampling: The Methods note that configurations drawn from a common seed share an optimization run, so the 30 pooled points are not fully independent. To put a number on that caveat rather than leaving it qualitative, we repeated the condition-number analysis with an exact permutation test that permutes *whole seeds* rather than individual configurations, preserving the within-seed structure. This is a post-hoc sensitivity analysis and was not preregistered.

The τ values are unchanged by construction; only the reference distribution differs. Under seed-level permutation the uncorrected p-values are 0.133 (Mahalanobis), 0.075 (cosine), 0.108 (kNN) and 0.025 (kernel density). None survives Holm correction across the four-estimator family, whose first threshold is 0.0125.

Two things follow, and they pull in opposite directions. The Holm-corrected significance reported above rests on treating the pooled configurations as independent draws, which the Methods already declined to claim; a reader who takes the seed as the resampling unit should not read Table I as establishing significance. But the permutation test is also very weak here: with five seeds its reference distribution admits only $5! = 120$ arrangements, so its p-values are quantised at multiples of $1/120 \approx 0.008$ and it has little power to resolve an association of any size. These p-values therefore record a failure to resolve, not a demonstration of absence, and we report both analyses rather than choosing between them.

In the companion analysis, probe accuracy was associated with LID ($\tau = 0.39$, $p = 0.005$) and with per-direction variance ($\tau = 0.48$, $p = 5.4\times10^{-4}$), and was not associated with spectral-decay slope ($\tau = -0.03$, $p = 0.85$) or with condition number ($\tau = -0.04$, $p = 0.77$).

D. Intervention Type and Geometry

All sixteen mixed-effects models converged. Three interaction terms reached Holm-corrected significance: LID with Mahalanobis ($p = 2.9\times10^{-6}$), LID with kernel density ($p = 2.6\times10^{-3}$), and condition number with Mahalanobis ($p = 3.9\times10^{-5}$). The remaining thirteen did not, including all four slope interactions and all four per-direction-variance interactions. Interaction coefficients for the Holm-significant terms were small in absolute magnitude (-1.2×10^{-4}, -4.3×10^{-2}, and 2.9×10^{-4} respectively).

E. Geometry Manipulation and the Causal Comparison

The normalization-based operator originally specified for the causal comparison was measured across all twenty Intervention-arm checkpoints before use. It moved condition number substantially (mean change -2.7×10^{5}) while leaving both preregistered metrics effectively unchanged: mean LID change -4×10^{-4} (SD 1.8×10^{-3}) and mean slope change below 10^{-6} (SD 10^{-6}). It was therefore not used.

The adopted fixed-exponent spectral tempering operator was subjected to the same check. Across five seeds it moved spectral-decay slope reproducibly and near-monotonically (-1.5×10^{-3} at $\gamma = 0.25$, -2.7×10^{-3} at $\gamma = 0.5$, -3.6×10^{-3} at $\gamma = 0.75$, -3.5×10^{-3} at $\gamma = 1.0$), and moved LID reproducibly but non-monotonically, rising by $+2.7$ to $+5.8$ at $\gamma \in \{0.25, 0.5\}$ before falling by -1.5 to -4.4 at $\gamma = 1.0$; the same inverted profile appeared in all five seeds. Gate 0 passed in all twenty post-transform configurations.

For the causal comparison, γ was calibrated on a single seed to bring the full-strength checkpoint's spectral-decay slope to the level occurring naturally at the weakest intervention strength, giving $\gamma = 0.01$ (slope 6.746×10^{-3} against a target of 6.751×10^{-3}). A single calibrated value rather than a ladder is used here because the comparison requires one specific condition—geometry matched to a reference checkpoint while nominal strength differs—rather than a dose–response relationship; the operator's dose behaviour is characterized separately by the manipulation check reported above. At this γ, LID remained 0.60 below the matched checkpoint's value: the comparison is therefore matched on spectral-decay slope, not on both preregistered metrics jointly. Three configurations were then evaluated across five seeds: full strength unmanipulated ($\alpha = 1.0$, $\gamma = 0$), weak strength unmanipulated ($\alpha = 0.25$, $\gamma = 0$), and full strength slope-matched to the

TABLE II
ESTIMATOR AUROC (MEAN OVER FIVE SEEDS) FOR THE CAUSAL COMPARISON. CONFIGURATION C IS MATCHED TO B ON SPECTRAL-DECAY SLOPE AND TO A ON NOMINAL INTERVENTION STRENGTH.

	A (α=1)	B (α=0.25)	C (α=1, γ=0.01)
Mahalanobis	0.9984	0.9984	0.9984
Cosine	0.9503	0.9750	0.9521
kNN	0.9928	0.9973	0.9928
KDE	0.7611	0.8713	0.7657

weak checkpoint ($\alpha = 1.0$, $\gamma = 0.01$). Gate 0 passed in all five manipulated configurations.

Table II reports estimator AUROC for the three configurations. Mahalanobis and nearest-neighbour showed no separation between any pair of configurations. Cosine and kernel density showed a separation between the two unmanipulated strengths (0.025 and 0.110 AUROC respectively), and on both, configuration C was numerically closer to its nominal-strength counterpart (A) than to its slope-matched counterpart (B).

None of the eight paired comparisons reached Holm-corrected significance. The smallest corrected p-value was 0.111 (nearest-neighbour, geometry-matched), and the smallest uncorrected p-values were 0.014 (nearest-neighbour, geometry-matched), 0.023 (cosine, nominal-strength-matched), and 0.048 (kernel density, geometry-matched); Wilcoxon signed-rank p-values were no smaller than 0.063 throughout. With five seeds and an eight-comparison correction family, statistical power was limited, and the comparison is therefore reported as inconclusive at the significance-test level.

V. DISCUSSION

A. Principal Findings

We tested whether an association reported across representations—between in-distribution geometry and distance-based estimator performance—can account for a previously reported phenomenon, the probe–reliability dissociation, when geometry is intervened upon within a single fixed frozen backbone. Under the evaluated backbone, layer, and task, that explanatory reading received limited empirical support. The previously reported dissociation was reproduced in direction with high statistical confidence, but at a magnitude approximately two orders of magnitude smaller than originally reported. Neither preregistered geometry metric showed a Holm-corrected association with estimator performance, and a preregistered comparison that matched spectral-decay slope across differing nominal intervention strengths did not reach corrected significance in either direction. Three of sixteen intervention-type-by-geometry interactions remained Holm-significant after conditioning on geometry which, under the preregistered criterion, is inconsistent with full mediation.

B. A Coherent Reading of the Four Analyses

The four analyses address different aspects of the question, and they are consistent with a single observation: on this backbone, distance-based estimators were already operating near ceiling before any intervention, with Reference-arm AUROC of 0.9984 for Mahalanobis and 0.9979 for nearest-neighbour. The domain shift studied here was, for these estimators, already highly separable.

This observation makes the remaining results easier to situate. Where an estimator has almost no headroom, there is correspondingly little variation in its performance for a geometry metric to covary with, which is consistent with the absence of corrected-significant associations for LID and spectral-decay slope. It is likewise consistent with the causal comparison, in which Mahalanobis and nearest-neighbour showed no separation between any configuration, and only the two estimators retaining measurable headroom— cosine and kernel density—separated at all. That two of the four estimators were insensitive to every configuration is itself a finding rather than an absence of one: it establishes that on this representation the two strongest distance-based detectors are unaffected by interventions of the magnitude studied here, which is directly relevant to whether geometry-targeted correction is warranted in deployment. And in the interaction models, where residual variation is at the fourth decimal place, a statistically detectable interaction term need not correspond to a substantively meaningful effect; the three Holm-significant coefficients were of order 10^{-4} to 10^{-2}.

We emphasize what this reading does and does not assert. It is a statement about the sensitivity our experiments could achieve, not about the account itself. An evaluation conducted where the phenomenon of interest is small provides limited power to test an explanation for that phenomenon; the appropriate conclusion is that this study offers limited empirical support for the account in this regime, not that the account is incorrect, nor that the geometric quantities it names are unimportant in regimes where the phenomenon is large.

C. Condition Number

Condition number was the one geometric quantity associated with all four estimators at corrected significance, while the two preregistered metrics were not. This study provides direct evidence bearing on how that association should be read. When we measured the normalization-based operator, it changed condition number by approximately 2.7×10^5 while leaving LID and spectral-decay slope unchanged to within 10^{-3} and 10^{-6} respectively. Condition number therefore responds strongly to a transformation that leaves both shape metrics untouched, consistent with it behaving substantially as a scale-sensitive descriptor rather than as a measure of the spectral shape at issue. This reading is reinforced by the operator's own specification, which adjusts feature radii while holding directions fixed [2]. Its association with estimator performance should accordingly not be read as support for the explanatory hypothesis, and we do not treat it as such.

D. Limitations

Several limitations bound these findings.

The correlational result depends on how the pooled configurations are treated. Holm-corrected significance for condition number holds when the 30 configurations are treated as the unit

of analysis; under an exact permutation test over whole seeds none of the four estimator associations survives correction. That sensitivity analysis is reported with the results and was not preregistered. The two analyses are not in conflict so much as differently powered: the seed-level test has only $5! = 120$ arrangements available and therefore very little resolution, so it cannot adjudicate an association of this size in either direction. What can be said is that the strength of the correlational claim is contingent on the resampling unit, and readers who take the seed as that unit should treat the association as suggestive rather than established.

The evaluation covers one backbone, one representation layer, one diagnostic task, and one domain-shift pair. That pair differs in imaging modality, not only in acquisition conditions: ISIC 2019 comprises dermoscopic images and PAD-UFES-20 close-up clinical photographs. A modality difference is a stronger form of distribution shift than the device- or site-level shifts that motivate distance-based reliability estimation in deployment, and it is the most plausible reason the two populations are as separable as we observe. Whether the near-ceiling estimator performance reflects a property of frozen foundation-model representations in general, or the modality gap between these particular populations, cannot be determined from this study. A same-modality shift—for instance between dermoscopic cohorts acquired at different sites or with different devices—would produce lower baseline separability and a correspondingly more sensitive test. We regard constructing such a comparison as a precondition for a stronger evaluation of the explanatory hypothesis in this regime rather than as a refinement of the present one.

The causal comparison was matched on spectral-decay slope alone. At the calibrated exponent, LID remained 0.60 below the target checkpoint's value, so the comparison does not establish matching on both preregistered metrics jointly. Because the account names both quantities, this is a genuine restriction on what the comparison can support.

The manipulation operator's effect on LID was reproducible but non-monotonic, rising at intermediate exponents before falling at full strength, with the same profile in all five seeds. A monotone dose–response was not assumed by the preregistered criterion, and the observed profile satisfies it; but it means the operator does not provide a simple one-directional LID control, which constrains the design of future comparisons.

Finally, five seeds with an eight-comparison correction family afford limited statistical power for the causal comparison, and we report it as inconclusive at the significance-test level rather than as evidence in either direction.

E. Implications

For practitioners deploying distance-based reliability estimation on frozen foundation-model representations, these results provide no evidence that geometry-targeted correction is required before such estimators can be used in the configuration evaluated here. This is a statement about the evaluated setting, and does not extend to configurations where baseline separability is lower.

Methodologically, the value of this study lies less in any individual statistic than in the sequence of checks that constrained them. Two elements proved decisive. The manipulation check identified, before any causal result was interpreted, that the operator originally specified for the comparison did not move the quantities at issue—a finding that would otherwise have surfaced only as an unexplained null. And preregistering the correction family before execution meant that several uncorrected p-values below 0.05 could be reported as what they are, rather than selected after the fact.

Rather than confirming an explanatory reading of the reported association, our results delineate the boundary under which such a reading can currently be supported: intervening on representation geometry within a single frozen backbone yielded limited empirical support for it, under the evaluated backbone, layer, and task.

VI. Conclusion

We tested whether an association reported across representations, between in-distribution geometry and distance-based estimator performance, can account for the previously reported probe–reliability dissociation when geometry is intervened upon within a single frozen dermatology foundation model. Under a preregistered protocol incorporating implementation-integrity verification and manipulation checking, that explanatory reading received limited empirical support in the evaluated configuration. The dissociation was reproduced in direction and with high statistical confidence, but at a magnitude approximately two orders of magnitude smaller than originally reported, and neither preregistered geometry metric showed a Holm-corrected association with estimator performance.

These findings indicate that explanatory accounts established in one architectural regime warrant direct evaluation before being carried into another. The protocol reported here—gating on implementation integrity, confirming that the manipulated quantity moves before any causal comparison is interpreted, and fixing the multiplicity correction in advance of execution—is specific neither to this account nor to this backbone, and provides a basis for assessing whether a previously reported effect, and the explanation advanced for it, hold under changed conditions.

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Preregistration and Reproducibility

Prior to any execution, a protocol document specifying the claims to be tested, the statistical test for each claim, the gate criteria, and the multiplicity-correction family was finalized and cryptographically checksummed. This document was not deposited with a public preregistration registry; the record is internal, timestamped, and available from the authors on request. All statistical analyses reported here follow that document, and deviations arising during execution were recorded as dated amendments rather than applied silently.

Result artifacts were frozen on completion, with per-file checksums recorded in a manifest, so that every number reported in this manuscript is traceable to an immutable file.

Data and Code Availability

Both datasets used in this study are publicly available from their original sources.

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